

Mathematics Extended Essay

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An investigation into the Euler Mascheroni constant and its
properties regarding convergence.

Research question: To what extent can the Euler-Mascheroni constant be generalized for the family of functions $f(x) = \frac{1}{a^x}$, $a > 1$?

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This investigation is an examination of some properties of the Euler-Mascheroni constant and how this relates to many simple properties in mathematics such as integration and summation. The Euler-Mascheroni constant originally defined by Leonhard Euler is given the notation γ . This constant is defined as the limit of the difference between the sequence of partial sums of the sequence $a_n = \frac{1}{n}$ and the sequence $b_n = \ln(n)$. Where “ln” is the logarithm in base e .

$$\gamma = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{1}{k} - \int_1^n \frac{1}{x} dx \right). \quad (1)$$

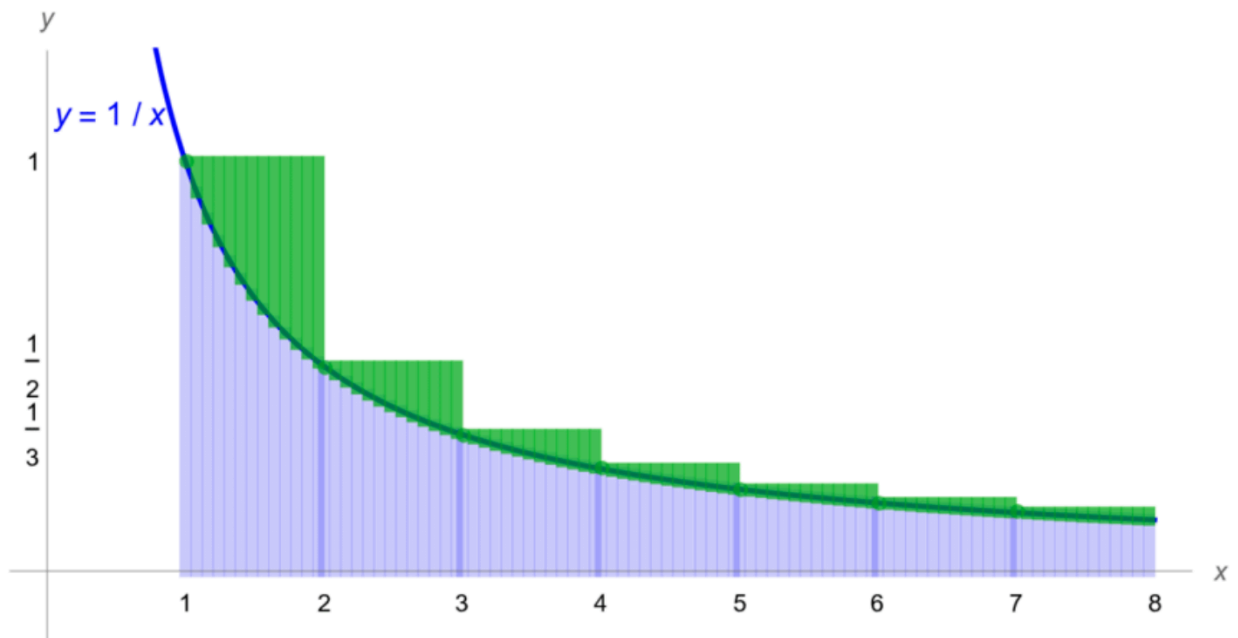


Fig. 1. Graph of Euler-Mascheroni constant. By Felix Martínez de la Rosa.

As shown in the graph of the Euler-Mascheroni constant (Figure 1) the difference between the area of the rectangles with a height of $\frac{1}{k}$ where k is the natural numbers and the area under the curve of $y = \frac{1}{x}$. In relation to the mathematical formula for the constant (1), the sum

of the rectangles are given by the sigma notation and the area under the curve is given in integral notation. The difference between the two areas is highlighted by the green area that will converge for the limit of x approaching infinity.

This constant is the difference between a summation and an integration which is quite interesting as both of them *diverge* to infinity but the difference of their limits is convergent. Figure 1 only shows up to $n = 8$ but of course the Euler-Mascheroni constant is equal to the limit as n approaches infinity. The underlying question being asked throughout the essay is regarding the generalization of this constant and if the difference between a summation and integration can exist as a constant similar to the Euler-Mascheroni constant (1). Whether or not a family of functions can *generalize* this original constant will depend on the difference of limits in the process of integration and summation. I will focus specifically on the family of functions $f(x) = \frac{1}{a^x}$, $a > 1$ and the research question logically follows as: **To what extent can the**

Euler-Mascheroni constant be generalized for the family of functions $f(x) = \frac{1}{a^x}$, $a > 1$?

Although the Euler-Mascheroni constant is not as ubiquitous in mathematics as other famous limits like Euler's number e or pi, it holds quite a lot of significance in the fields of complex and transcendental number theory and still is relevant to mathematicians today. This constant is often denoted by a lowercase gamma (γ) because of its connection to the gamma function but in the times of the famous Leonhard Euler this constant was denoted as C and the symbol gamma was "first used by Mascheroni (1790)" (Weisstein) which is why the two mathematicians share the name of this constant today. For the purposes of this essay the Euler-Mascheroni constant will be denoted as a lowercase gamma " γ " and the generalized constant will be denoted as a capital C .

In my research I took the classical Euler constant and generalized it for a family of functions in order to investigate the properties of these constants and understand their behaviors in relation to the original Euler constant.

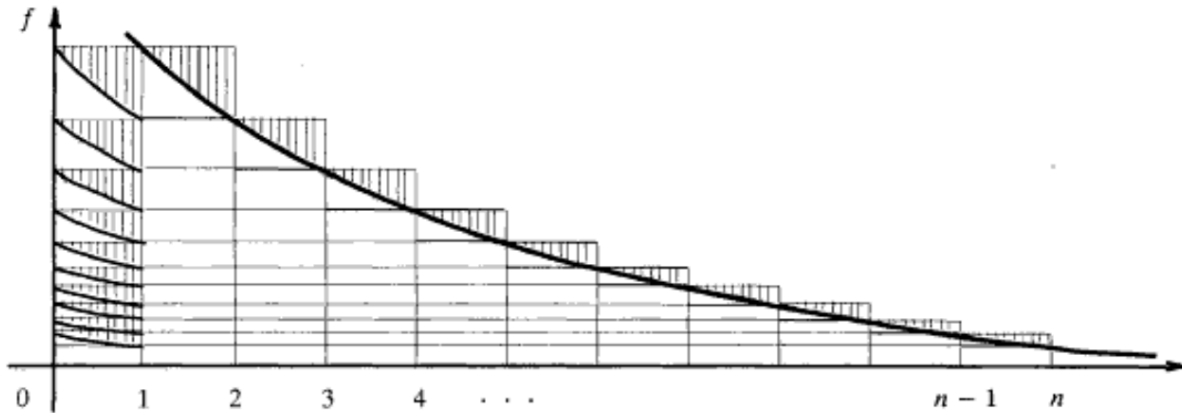


Fig 2. The visual representation of a generalized Euler-Mascheroni constant. Tom Apostol.

The visual representation of a generalized Euler-Mascheroni constant as shown in Figure 2 helps illustrate this difference where the shaded pieces are compared to the function itself. In this version Apostol does not use the Euler-Mascheroni constant but instead a generalization similar to the family of functions used in my research. What is important to note is that Apostol similarly uses the limit as n approaches infinity to talk about the Euler-Mascheroni constant which again is convergent but can be expressed as the difference of the limit between two divergent functions.

On the harmonic sequence and series

The harmonic sequence and series gets its name from Greek philosopher and mathematician Pythagoras who is thought to have discovered the harmonic mean which is what

creates the harmonic sequence and series (Kullman). The harmonic mean of any two real numbers a and b is equal to $\frac{2ab}{a+b}$ (Kullman), which is what gives way to the harmonic

progressions later on. Consider two sequences: $a_n = a_1 + d(n - 1)$, $n \in \mathbb{N}$, $a_n \neq 0$, and

$b_n = \left\{ \frac{1}{a_1}, \frac{1}{a_1+d}, \frac{1}{a_1+2d}, \dots \right\}$, $n \in \mathbb{N}$, where b_n is the sequence of the reciprocal values of a_n . We

define b_n as the harmonic *progression* of the sequence a_n . If the sequence a_n is equal to the sequence of natural numbers $\mathbb{N}$ then the sequence b_n is a special case called the harmonic sequence, i.e. $b_n = \left\{ \frac{1}{n}, \frac{1}{n+1}, \dots \right\}$, $n \in \mathbb{N}$.

Theorem: Let H be the harmonic mean of two real numbers a, b . If there exists a sequence b_n such that b_n is the harmonic progression of an arithmetic sequence a_n and a, b are both terms in this sequence: $a = \frac{1}{(a_1+id)}$ and $b = \frac{1}{(a_1+(i+2)d}$, $i \in \mathbb{N}$, then the term which is inbetween a, b in b_n , i.e. $\left(\frac{1}{a_1+(i+1)d} \right)$, must be equal to H .

Proof: Recall that the harmonic mean of two numbers is equal to $\frac{2ab}{a+b}$, by substituting the values of a and b into this equation the result is as follows: $\frac{2 \cdot \left(\frac{1}{(a_1+id)} \right) \cdot \left(\frac{1}{(a_1+(i+2)d)} \right)}{\left(\frac{1}{(a_1+id)} \right) + \left(\frac{1}{(a_1+(i+2)d)} \right)}$. Using the

sequence of harmonic progressions $b_n = \frac{1}{a_n}$, $n \in \mathbb{N}$. It is clear that, as justified in the theorem,

there exists a term in the sequence which is in between a and b . By algebraic manipulation the

fraction above of H can be turned into $\frac{\frac{2}{(a_1+id)(a_1+(i+2)d)}}{\frac{2a_1+2id+2d}{(a_1+id)(a_1+(i+2)d)}}$, which is then equal to $\frac{1}{a_1+id+d}$ and going

back to the theorem it was expected that H be equal to the term in between a and b which is

$\frac{1}{a_1 + (i+1)d}$. Hence it is proved that the harmonic mean H of any two numbers a, b will be the middle term in a harmonic progression b_n . ■

For the purpose of this essay there is not any evaluation done about the number classification of the Euler constant but simply a question of whether or not these constants can be generalized at all. This is done because the classification of the Euler-Mascheroni constant has not yet been proved as either transcendental or algebraic and this difference is not very relevant to the research. Although this sequence does have its roots embedded in history it is quite relevant to mathematicians even today and for one it is an example of a series which is both *monotone* and decreasing towards 0, yet divergent. The term monotone means either the term in a sequence of terms a_n is always greater than the next term a_{n+1} or that the next term a_{n+1} is always greater than the previous term a_n . For all terms in a sequence a_n $n \in \mathbb{N}$ must be either $a_n > a_{n+1}$ or $a_n < a_{n+1}$ for it to be considered monotone. Regarding a series' divergence, it will only diverge to infinity if there is no possible output value which cannot be exceeded by some arbitrarily large value of n . In the case of the Euler-Mascheroni constant, both the function of the natural logarithm of x and the series of $\sum_{n=1}^{\infty} \frac{1}{n}$ are divergent to infinity.

Theorem The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent to infinity.

Proof: $\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} + \dots,$

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \left(\frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \frac{1}{14} + \frac{1}{15} + \frac{1}{16}\right) + (\dots) .$$

$$>$$

$$1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16}\right) + (\dots) =$$

$$1 + \frac{1}{2} + \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) + (\dots).$$

We notice that the series $1 + \frac{1}{2} + \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) + (\dots)$ is divergent; therefore, the harmonic series will be divergent as it is strictly greater than the divergent series

$$1 + \frac{1}{2} + \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) + (\dots). \blacksquare$$

We may now move toward the integral of $\frac{1}{x}$ which is much more simple. Recall that the indefinite integral expresses the area under a curve for a function. Our function $\frac{1}{x}$ will continue for x approaching infinity, this means the curve will always have an area underneath it and such can never be equal to 0. By logical deduction since the integral will always be greater than 0 for any value of x it means we can confirm that the $\int_1^{\infty} \left(\frac{1}{x}\right)$ will be divergent to infinity.

The graph of this series can be represented in figure 2 shown below which demonstrates how the divergence process is quite slow, for instance it takes $(1.5 \cdot 1043)$ terms to have a partial sum which is greater than 100.

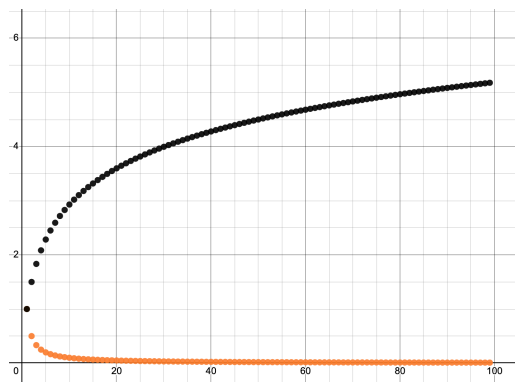


Fig. 3. Partial sums of the harmonic series and sequence up to 100 terms. Made by candidate using the web program Desmos.

In figure 3 a graph is shown of the partial sums of the harmonic sequence and series, the black points indicate the sequence of the partial sums of the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ where the x-axis corresponds to the value of n and the orange dots represent the harmonic sequence $b_n = \frac{1}{a_n}$, $n \in \mathbb{N}$. The harmonic sequence is monotone and decreases towards 0 as shown by the orange points in figure 2.

Integration and summation

Since this specific sequence deals heavily with the topic of limits and this is not extensively studied in the curriculum, it is also very relevant to include a specific proof about the nature of limits and what it means for a series or sequence to be approaching a limit. In words, the limit can simply be described for a sequence a_n as follows: for any measure of closeness ϵ to the limit a there is always a value of the sequence n which is closer to the limit than ϵ . For the specific case of the Euler-Mascheroni constant and the generalized constants, both are always positive, meaning that only the limit approaching from positive values need to be considered. In a mathematical sense the limit can be shown as:

$$\forall \epsilon > 0, \exists N \in \mathbb{N}, \text{ such that for any } n \geq N, |x_n - x| < \epsilon \quad (1)$$

Whereas in this case as stated earlier the absolute value is not explicitly necessary because the difference between $x_n - x$ will always be positive given the parameters of the sequences considered in this essay.

In unison with this theorem regarding a limit of a sequence it is important to also present the idea of a monotone sequence and introduce the monotone sequence theorem. This theorem

states that a *bounded* monotone sequence will have a limit, thus it must also be convergent. A bounded sequence is any sequence a_n which is bounded both above and below by constants B, and C:

$$C \leq a_n \leq B. \text{ for all values of } n.$$

The Monotone Sequence theorem states that: “If a_n is bounded and a_n is also monotone, then it must be convergent.” (Davies). Recall that a monotone sequence is one which either strictly decreases or increases is called monotone.

Euler-Mascheroni constant

Theorem: The Euler-Mascheroni constant or $\lim_{n \rightarrow \infty} (\sum_{k=1}^n \frac{1}{k} - \ln(n))$ is both monotone

and bounded; by the monotone sequence theorem the limit must exist.

Proof: $\gamma = \lim_{n \rightarrow \infty} (\sum_{k=1}^n \frac{1}{k} - \int_1^n \frac{1}{x} dx); \int_1^n \frac{1}{x} dx = \ln(n)$, using the formula for the

Euler-Mascheroni constant noted previously we can use the integral notation of the natural log to begin the proof. In the following proof it will be proved that this constant is less than 1.

$$\lim_{n \rightarrow \infty} (\sum_{k=1}^n \frac{1}{k} - \int_1^n \frac{1}{x} dx) = \lim_{n \rightarrow \infty} (\sum_{k=1}^n \frac{1}{k} - \sum_{k=1}^{n-1} \int_k^{k+1} \frac{1}{x} dx) \quad (2).$$

In (2) we use the property of an integration which is that it can be rewritten as the summation of each individual portion of the integration using the same parameters as before, now expressed as a summation.

$$= \lim_{n \rightarrow \infty} (\sum_{k=1}^{n-1} \frac{1}{k} + \frac{1}{n} - \sum_{k=1}^{n-1} \int_k^{k+1} \frac{1}{x} dx) \quad (3).$$

Now we split the summation of the harmonic series into the summation of $n-1$ terms and the final term added on the end, it is becoming clear now that these two summations can be combined

which will happen in (4).

$$= \lim_{n \rightarrow \infty} \left(\sum_{k=1}^{n-1} \left(\frac{1}{k} - \int_k^{k+1} \frac{1}{x} dx \right) + \frac{1}{n} \right) \quad (4).$$

$$\leq \lim_{n \rightarrow \infty} \left(\sum_{k=1}^{n-1} \left(\frac{1}{k} - \frac{1}{k+1} \right) + \frac{1}{n} \right) \quad (5).$$

Using the knowledge that the function $f(x) = \frac{1}{x}$ is decreasing for increasing values of x , we understand that the upper boundary for the integral must be less than the lower boundary. Hence

$$\frac{1}{k+1} \leq \frac{1}{k} \text{ which is true for the } \lim_{k \rightarrow \infty} \frac{1}{k}.$$

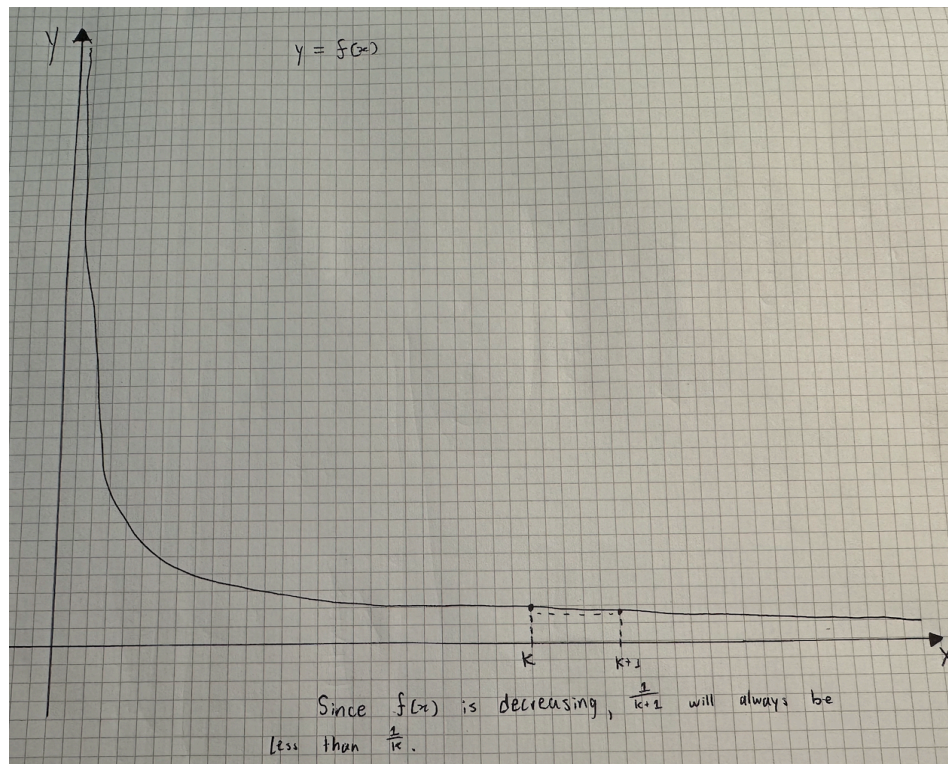


Fig. 4. Visual representation of a standard integral on a decreasing function. Made by candidate.

As shown in Figure 4, the unit interval and area from $\frac{1}{k}$ to $\frac{1}{k+1}$ is less than the integral and in (4) this integral is being subtracted, thus by taking a value which must be less than this integral we can confirm that the result in (5) is larger than (4).

$$= \lim_{n \rightarrow \infty} ((1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + (...) + \frac{1}{n}) = 1 \quad \text{Therefore}$$

$$\lim_{n \rightarrow \infty} (\sum_{k=1}^{n-1} (\frac{1}{k} - \int_k^{k+1} \frac{1}{x} dx) + \frac{1}{n}) \leq 1 \quad (6)$$

Now that we have proved that the Euler-Mascheroni constant is less than 1, in order to prove that it is greater than 0 we must conduct another proof. Similarly, to prove that γ is also always increasing and thus monotone will also be done, from the outset we define that a_n is equal to the Euler-Mascheroni constant.

$$0 < a_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} - \ln(n) < 1$$

a_n will be convergent if it can be proven that the sequence is constantly increasing and greater than 0, thus to prove that a_n is greater than 0 we can simply use a graphical analysis on a calculator.

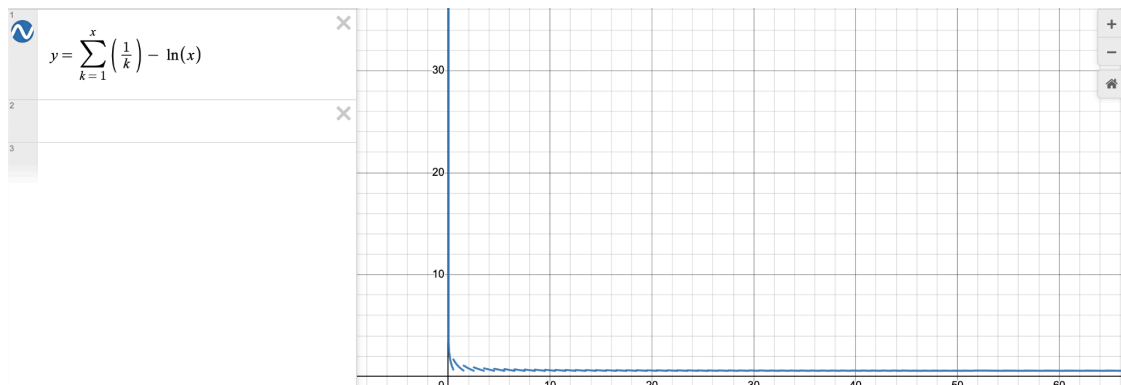


Fig. 5. Graphical representation of the function shown above in (6). Made by candidate using Desmos.

In the graph shown in Figure 5 we see that the harmonic series minus the natural logarithm of x is in fact always greater than 0 (desmos). The next step in this process is to show that a_n will be monotone.

$b_n = a_n - \frac{1}{n}$, here we define a_n by another sequence which we call b_n . This new function will converge ($-1 < b_n < 1$) as long as:

$$b_{n+1} > b_n, \text{ therefore this is true if and only if } b_{n+1} - b_n > 0 \quad (8)$$

$$b_{n+1} - b_n = ((1 + \frac{1}{2} + \dots + \frac{1}{n-1}) + \frac{1}{n} - \ln(n+1)) - ((1 + \frac{1}{2} + \dots + \frac{1}{n-1}) - \ln(n)) \quad (9)$$

$$= \frac{1}{n} - \ln(n+1) + \ln(n) \quad (10)$$

$$= \frac{1}{n} - \int_n^{n+1} \frac{1}{x} dx > \frac{1}{n} - \int_n^{n+1} \frac{1}{n} dx;$$

In this step we simplify the natural log into the integral and begin to show how it will be greater than its other component.

$$\text{Since } \frac{1}{n} - \int_n^{n+1} \frac{1}{x} dx = 0 \quad (11) \text{ then}$$

$$b_{n+1} - b_n > 0 \quad (12)$$

Since b_n is always increasing then this also satisfies the original expression that $a_n - \frac{1}{n}$ must also be increasing. The monotone sequence theorem suggests that given a_n is always increasing but bounded above it must be convergent (Hamming). ■

Generalization of the Euler-Mascheroni constant

Now that we have proved the Euler-Mascheroni constant to be convergent we can begin to look at the generalization of these constants. One researcher, Tom Apostol who created a

previous graph used in this essay gives the proof for any generalized Euler constant which can exist in the following form:

$$C(f) = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n f(k) - \int_1^n f(x) dx \right) \quad (1)$$

Where f is a positive and strictly decreasing function on $[1, \infty)$ (Apostol). One example of this constant is where $f(x)$ is equal to $\frac{1}{x}$ which will give the Euler-Mascheroni constant γ . This general formula is derived from Euler's original method of finding the constant which is known as "Euler's summation formula" (Apostol) or sometimes also known as the Euler-Maclaurin summation formula, which essentially geometrically compares the integration for a decreasing function and a series which approximates the area under the function with rectangles, comparing the inherent difference between the two and finding a constant. As shown before in Figure 1 this difference can also be represented visually by using the area of rectangles as well as integrals. The important difference between the way Euler defined his constant and Apostol's formula for a general one is that these generalized Euler constants can now be thought of as the difference between the area under a curve $f(x)$ and a series with the same function $f(k)$. The inherent difference between the two can be quantified and visualized as such by comparing them geometrically. Hence the original hypothesis for the essay follows:

Theorem: Given the family of functions $f(x) = \frac{1}{a^x}$, $a > 1$, the generalized

Euler-mascheroni constant is $C(f) = \left(\frac{1}{a-1}\right) - \left(\frac{1}{a \cdot \ln(a)}\right)$

Proof: First we will prove this theorem for $a=2$ using (1):

$$C(\frac{1}{2^x}) = \lim_{n \rightarrow \infty} (\sum_{k=1}^n (\frac{1}{2^k}) - \int_1^n (\frac{1}{2^x}) dx) \quad (2)$$

This equation above can be simplified further which obtains the following:

$$C(\frac{1}{2^x}) = (\frac{0.5}{1-0.5}) - \int_1^n (2^{-x}) dx \quad (3)$$

Equation (3) follows logically from the equation (2), remember that the summation of a geometric series to infinity can be approximated by the formula $\frac{T_1}{1-r}$; in which T_1 is the first term in the sequence and r is the rate at which the sequence changes going from one term to the next. Hence, the equation (3):

$$C(\frac{1}{2^x}) = 1 - \frac{1}{2 \cdot \ln(2)} \quad (4)$$

To go from equation (3) to equation (4) we must evaluate the integral $\int_1^n (2^{-x}) dx$, using the method of substitution, a variable can be substituted in replace of “-x” where solving for this substituted variable results in an integrand in the form $-\frac{1}{\ln(2)} 2^u$ (where “u” is the variable being used to substitute). Substituting u back into the result gives the solution: $-\frac{1}{2^x \cdot \ln(2)}$. The constant at this point can easily be computed and evaluated at 4 significant figures; the result is ≈ 0.279 , where we substitute the upper and lower bound value into the function as such:

$(-\frac{1}{2^n \cdot \ln(2)}) - (-\frac{1}{2^1 \cdot \ln(2)})$. This decimal value is not of much interest, however it does go in

accordance with the Euler-Mascheroni constant. The function used for the generalized constant is in fact not divergent which means the difference between the integration and summation of said

function $f(x) = \frac{1}{2^x}$ is less than that of $f(x) = \frac{1}{x}$, which is the Euler-Mascheroni constant.

Now that one function in the family of functions has been used to generalize the Euler constant, it seems as if the Euler-Mascheroni constant can in fact be generalized to a great extent by this family of functions following from the original where $a=1$ and increasing from there. The main difference between the original Euler constant and the new generalized constants are the divergence of the series, obviously the harmonic series is divergent but the series used to calculate the generalized constants are convergent using the calculation from equation (3). In this sense the new generalized constants are instead not similar to the Euler-Mascheroni constant given that this difference is of finite values as opposed to the ones used in the original constant which are both infinite. Note though that the series itself not being divergent does not really change the nature of the actual constant in relation to the Euler constant, but it instead becomes a much more simple generalization because both the series and the integration can be calculated. Meaning that there is still an inherent difference geometrically speaking between the integration and the summation which can be demonstrated by both the original Euler-Mascheroni constant as well as the generalized ones.

From the result of equation (4) the logical conjecture can be made that as the series reaches infinity for a convergent series, the integration will be an increasingly more accurate approximation of the series, thus for functions which *converge* at a faster rate the generalized constant will become smaller. The word “converge” used above is relating to the fact that the function $f(x) = \frac{1}{3^x}$ will reach its asymptote with less values of x than in the function $f(x) = \frac{1}{2^x}$. In order to further describe this theory there needs to be a way of showing that

$C(f)$ will in fact tend towards 0 but not go beyond that, the simplest way of doing so is by solving the equation (1) for a and finding a relation which describes the function for any value a which will give a faster computational method for these constants $C(f)$.

$$\begin{aligned}
 C\left(\frac{1}{a^x}\right) &= \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \left(\frac{1}{a^k}\right) - \int_1^n \frac{1}{a^x} dx \right) \\
 C\left(\frac{1}{a^x}\right) &= \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \left(\frac{1}{a^k}\right) - \int_1^n a^{-x} dx \right) \\
 C\left(\frac{1}{a^x}\right) &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{1}{a^k}\right) - \left(\frac{1}{a \cdot \ln(a)}\right) \\
 C\left(\frac{1}{a^x}\right) &= \left(\frac{1}{a-1}\right) - \left(\frac{1}{a \cdot \ln(a)}\right) \quad (6) \blacksquare
 \end{aligned}$$

Now with this final result for the generalized constants it is clear that the theory previously stated is correct as the first term $\left(\frac{1}{a-1}\right)$ is greater than the second term $\left(\frac{1}{a \cdot \ln(a)}\right)$ which means subtracting the second from the first will never yield a result less than 0. And again the term $\left(\frac{1}{a \cdot \ln(a)}\right)$ cannot reach 0 as a is a real number, neither can the term $\left(\frac{1}{a-1}\right)$ which means that the constant should never reach 0 but asymptotically approach it as a tends towards infinity.

Conclusion

It is already clear that the Euler constant can be generalized to a great extent by the family of functions $\frac{1}{a^x}$ using Apostol's formula but it is important to note that these constants $C(f)$ are also proof of the theory that there will always be a difference between the infinite series

and the area under a curve. This inherent difference stems directly from the theory behind an integration, given that we begin thinking about the integral as the infinite summation of areas under a line which can be approximated by many rectangles with an arbitrarily small width. This means that by comparing the integration to the series we are thinking about having a unit distance between each rectangle and then adding all of the areas up, and contrasting this idea with taking an upper and lower boundary and using an infinite amount of rectangles to find an area under the curve. This inherent difference between the two ideas is highlighted by this research because the constant essentially proves that this difference exists, not only that but the constant also gives important context to infinity and how mathematicians think about the meaning of infinity. Namely, the Euler-Mascheroni constant is the limit of the difference whereas the generalized constants are the difference between limits, although it is quite confusing it is also very philosophically interesting in the context of calculus and how infinity is thought of depending on the situation.

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